Biochemistry Cumulative Exam – February 15, 2013

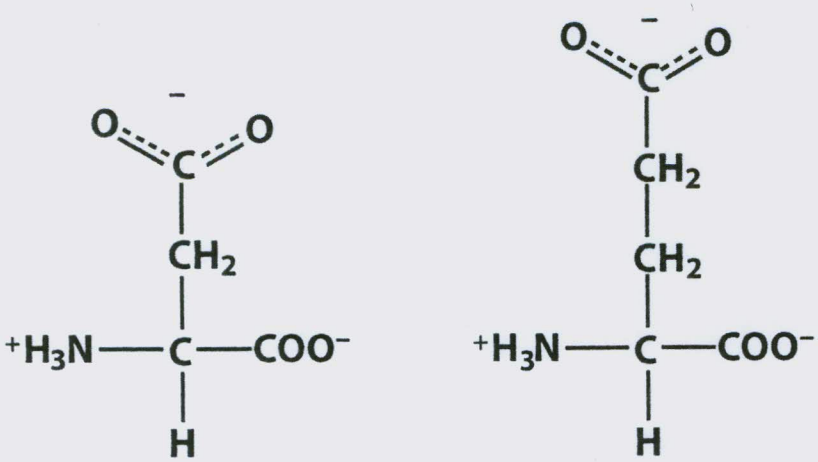
(Supplemental Information Attached)

A mouse is fed a diet including the amino acid aspartate, but no carbohydrate or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

A) Give a metabolic pathway for the synthesis of the amino acid glutamate using aspartate as the sole source of carbon and nitrogen. Briefly discuss the key reactions and enzymes and coenzymes required.

B) Write a complete, balanced equation for the pathway in A above, giving the stoichiometries of all molecules consumed and generated.

C) If aspartate labeled with C14 at carbon 3 is added to mitochondria, which carbon(s) of glutamate will become labeled?



Aspartate (Asp, D) **Glutamate (Glu, E)**

Figure 3.7
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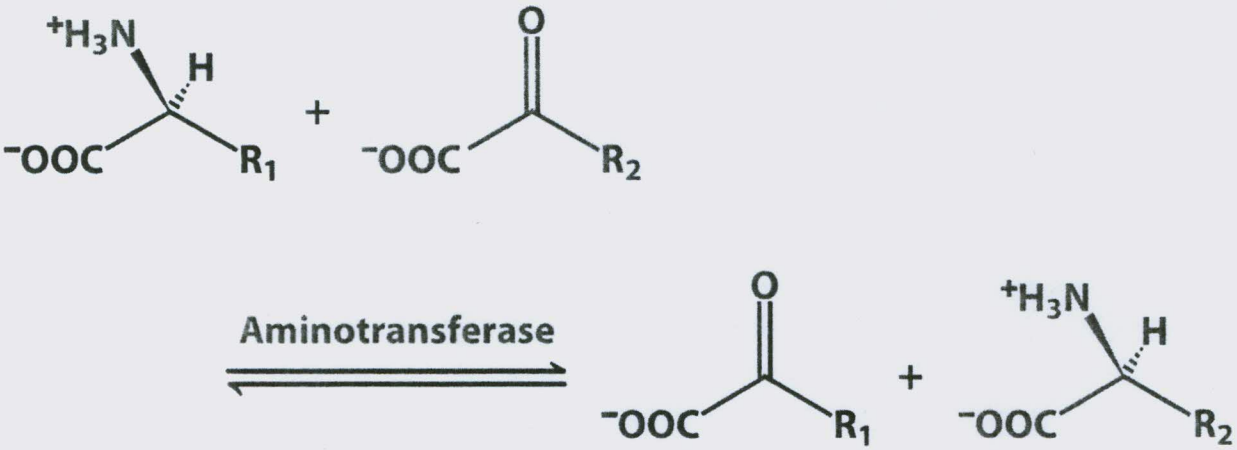

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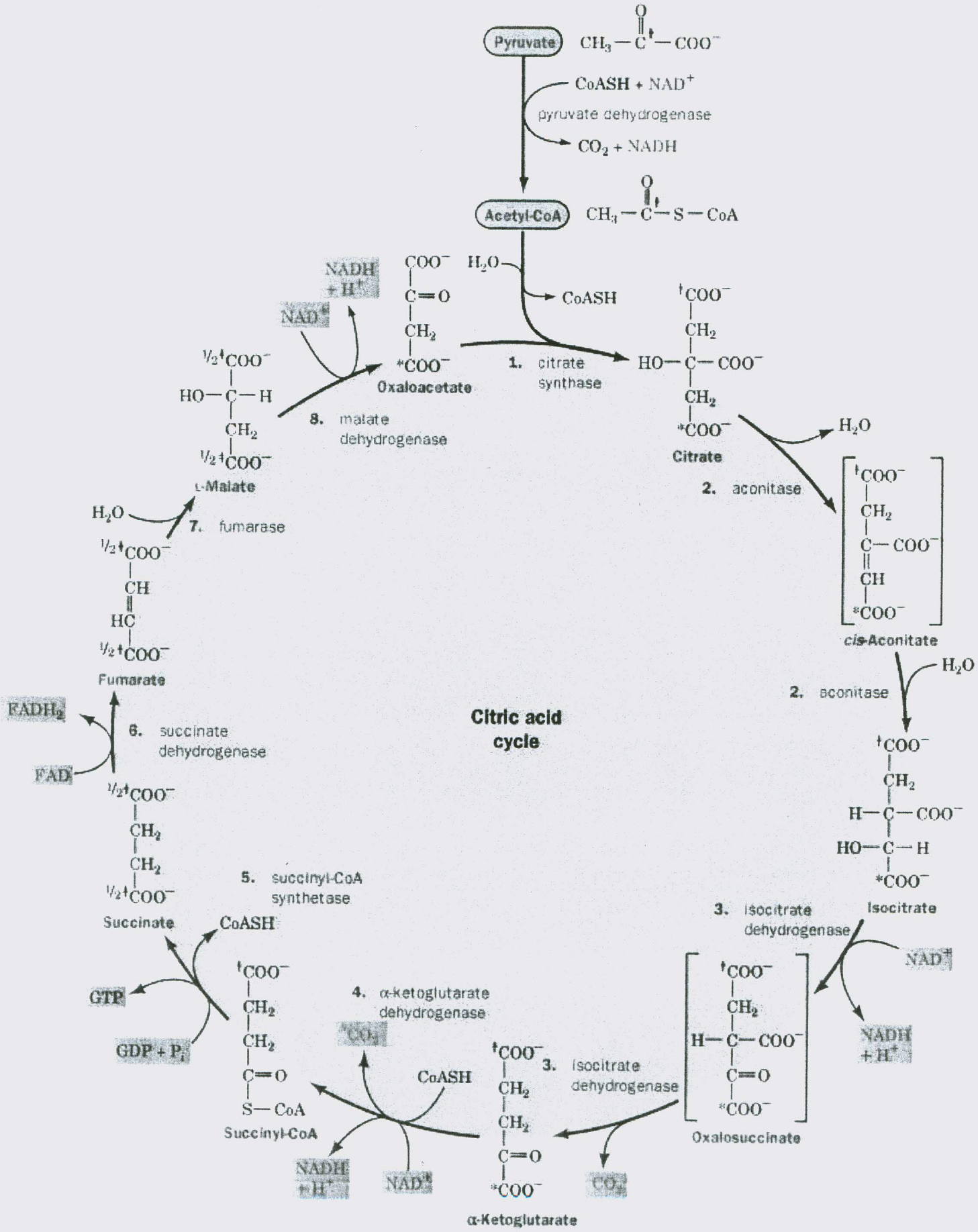
Pyruvate
CoASH + NAD+
pyruvate dehydrogenase
CO2 + NADH
Acetyl-CoA
H2O
CoASH
1. citrate synthase
Oxaloacetate
Citrate
2. aconitase
H2O
cis-Aconitate
2. aconitase
H2O
Isocitrate
3. isocitrate dehydrogenase
NAD+
NADH + H+
Oxalosuccinate
3. isocitrate dehydrogenase
CO2
α-Ketoglutarate
4. α-ketoglutarate dehydrogenase
CO2
CoASH
NADH + H+
NAD+
Succinyl-CoA
5. succinyl-CoA synthetase
CoASH
GTP
GDP + Pi
Succinate
6. succinate dehydrogenase
FADH2
FAD
Fumarate
7. fumarase
H2O
L-Malate
8. malate dehydrogenase
NADH + H+
NAD+
Citric acid cycle